Quantum Computing: Architecture and Applications

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Abstract

This research paper examines superconducting quantum processors and fault-tolera nt quantum error correction. The primary objective is to scale quantum computing beyond 10,000 physical qubits. Our experimental prototype, codenamed Helios-X, achieved an average 2-qubit gate fidelity of 99.85% under cryogenic temperatures of 15 millikelvin. The paper outlines the hardware architecture, quantum error decoding algorithms, and empirical benchmarks in chemical molecular simulation.

1. Hardware Architecture

The Helios-X processor integrates 1,024 transmon qubits coupled via tunable capa citive resonators. The dilution refrigerator maintains an operating base tempera ture of 15 mK with cooling power of 1.2 milliwatts. Signal readout is performed by Josephson parametric amplifiers operating between 4 GHz and 8 GHz. Total powe r consumption for the auxiliary cryostat systems is 14.5 kilowatts.

2. Experimental Benchmarks and Key Findings

The main findings demonstrate a 120-fold quantum speedup over classical supercom puters in simulating ground-state molecular orbitals of caffeine (C8H10N4O2). Th e surface code decoder reduced logical error rates to 1.2e-6 per cycle. However, coherence time T1 remains limited to 185 microseconds due to dielectric substra te losses.

3. Known Limitations and Future Work

The major limitation of the current architecture is cross-talk noise during simu ltaneous 2-qubit gates. Furthermore, the microwave wiring infrastructure generat es thermal load bottlenecks above 2,000 qubits. Future work will focus on 3D int erconnects and optical interconnect interfaces.